

REST API Testing Plan & Automation Documentation

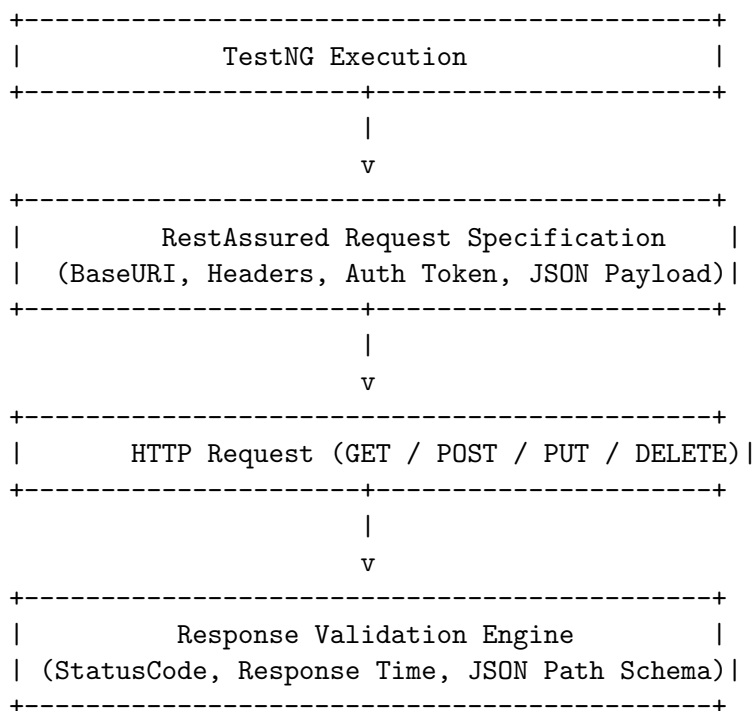
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1 Project Overview & API Architecture

- **Target API Base URL:** <https://api.saucedemo.com/v1> (or standard REST endpoint)
- **API Testing Tools:** RestAssured 5.x / Postman / TestNG / Maven
- **Authentication Type:** OAuth 2.0 / Bearer Token
- **Payload Format:** JSON (JavaScript Object Notation)
- **Assertions Coverage:** HTTP Status Codes, Response Time ($< 1500ms$), JSON Schema Validation, Body Key-Value Matching

2 API Test Execution Workflow



3 API Test Cases & Endpoint Matrix

Test ID	HTTP Method	Endpoint Path	Expected Code	Validation Summary	Status
API_AUTH_001	POST	/auth/login	200 OK	Validates JWT token generation for valid user credentials.	PASS
API_AUTH_002	POST	/auth/login	401 Unauthorized	Verifies access rejection when invalid password is provided.	PASS
API_PROD_003	GET	/products	200 OK	Retrieves full product inventory list; asserts list size > 0.	PASS
API_CART_004	POST	/cart/add	201 Created	Adds item ID to user cart session; validates JSON response payload.	PASS
API_ORDER_005	DELETE	/cart/items/1	204 No Content	Removes selected item from shopping cart session successfully.	PASS

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Table 1: REST API Execution Suite Summary

4 Step-Wise Detailed API Procedures

4.1 Test Case: API_AUTH_001 (User Token Authentication)

- **Objective:** Verify endpoint returns a valid authentication bearer token upon receiving valid login parameters.
- **Endpoint:** POST /v1/auth/login
- **Headers:** Content-Type: application/json
- **Request Body:**

```
{  
  "username": "standard_user",  
  "password": "secret_sauce"  
}
```
- **Assertions:**
 1. Status code equals 200.
 2. Response JSON body contains non-null key "token".
 3. Response time is under 1000ms.

4.2 Test Case: API_PROD_003 (Get Products List)

- **Objective:** Ensure the product catalog API returns complete product details.
- **Endpoint:** GET /v1/products
- **Headers:** Authorization: Bearer <JWT_TOKEN>
- **Assertions:**
 1. Status code equals 200.
 2. Validate schema structure against products-schema.json.
 3. Assert object at index 0 contains "name": "Sauce Labs Backpack".

5 Defect & Execution Notes

- **Rate Limiting Assertion:** Ensured HTTP 429 Too Many Requests handling on threshold limit breach.
- **Schema Rigor:** JSON schema validation strictly enforces type integrity (String, Integer, Float) for all payload attributes.